

A Semisupervised Aircraft Fuselage Defect Detection Network With Dynamic Attention and Class-Aware Adaptive Pseudolabel Assignment

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Abstract—To track the problem of aircraft fuselage defect detection in complex environments and reduce aviation safety hazards such as careless observation and delayed reporting due to objective factors, a semisupervised aircraft fuselage defect detection network was proposed. First, we constructed a new baseline model that extends one-stage detector with dynamic head and partial convolution named as dynamic decoupled detector, which enhances the representation capability of the model and improves the detection accuracy of small defects. Second, to address the issue of inconsistent pseudolabel distribution in semisupervised learning, we propose a class-aware adaptive pseudolabel assignment strategy that adaptively obtains the pseudolabel filtering threshold during the training iteration to further optimize the pseudolabel assignment process. Finally, to validate the effectiveness of the proposed model, we construct a dataset for aircraft fuselage defect detection for semisupervised training. Experimental results show that the proposed semisupervised aircraft fuselage defect detection network outperforms the current state-of-the-art semisupervised object detection framework on the aircraft fuselage defect dataset. At the same time, the proposed model has better generalization performance and provides more reliable support for real-time visualization of aircraft fuselage defects.

Impact Statement—Although the existing semisupervised target detection algorithm has excellent performance, there are still difficulties and challenges in aircraft fuselage defect detection. On the one hand, the algorithm proposed in this article can solve the relatively low detection accuracy of small- and medium-sized defects in aircraft fuselage defect detection to a certain extent; on the other hand, it can solve the label mismatch problem in industrial datasets and optimize the pseudolabel assignment process in semisupervised learning. Using our model not only improves the accuracy of aircraft fuselage defect detection but also saves the time of manual sample labeling to a great extent.

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Index Terms—Defect detection, pseudolabel assignment, semisupervised learning.

I. INTRODUCTION

AS we all know that the fuselage is one of the most critical components of an aircraft. The safety inspection of aircraft fuselage is a crucial and complex task in the process of aircraft quality inspection, and its safety plays an indispensable role in maintaining aviation safety. The fuselage withstands extreme pressure, elevated temperatures, and all kinds of impacts. Defects in aircraft fuselage, including scratch, rust, rivet-damage, and paint-peel, are caused and created by harsh working environments. Defects that are not addressed in a timely manner can be further exacerbated and pose a serious threat to the safe operation of the aircraft. Fuselage damage-induced catastrophic and unforeseen incidents pose a significant risk to the security of aircraft operations. Hence, it is imperative to conduct routine examinations of the fuselage in order to guarantee its well being and dependability.

In recent years, the increasing popularity of deep learning has led to the extensive utilization of convolutional neural networks in defect detection models within industries, owing to their exceptional capabilities in extracting features. However, for aircraft fuselage defect detection, defect detection models based on supervised learning convolutional neural networks require a substantial quantity of annotated data for training. Aircraft fuselage defects are complex and need to be labeled by specific experts [1]. Therefore, it is imperative to devise a framework for fuselage defect detection that can be trained using unlabeled data, as the availability of defect samples is limited, and annotating them requires significant time and effort. To facilitate the training of models with substantial quantities of unlabeled data, several semisupervised learning approaches have been suggested. Among these, the predominant method leverages abundant unlabeled data by employing an automated pseudolabel generation technique instead of relying on manual labeling. While semisupervised learning-based object detection has made great progress [2], [3], [4], there are still two key issues to introduce it into the detection of defects in aircraft fuselage.

First, in the field of defect detection, there has been a growing interest in unsupervised learning-based detectors in recent times

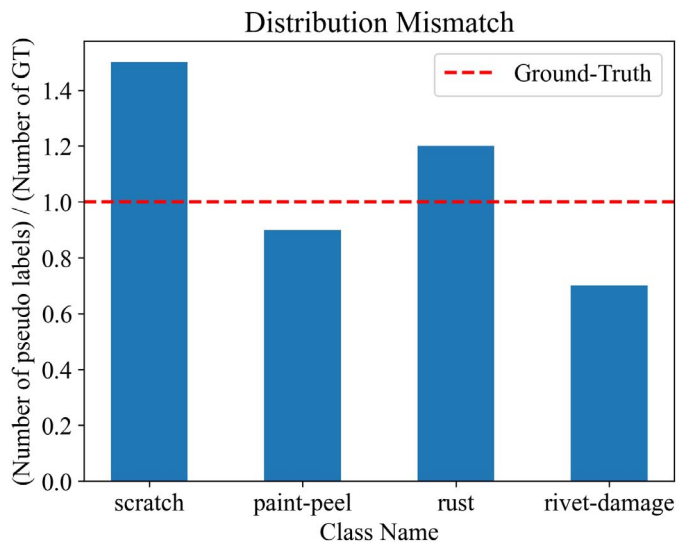


Fig. 1. Label data distribution mismatch problem on aircraft fuselage dataset. In the process of training, there is a bias between pseudolabels generated by a single confidence threshold and ground-truth labels (GT), as shown by the relationship between the blue bar and the red dashed line.

[5], [6], [7]. While one-stage detectors based on semisupervised learning exhibit benefits such as excellent recall, robust numerical stability, and efficient training speed, they are extensively employed in industrial detection scenarios that demand high recall. However, the majority of defect detection frameworks based on semisupervised learning have encountered challenges in achieving high accuracy in detecting defects and effectively identifying tiny defects. For the detection of fuselage defects, there are a large number of tiny defects, and the diversity of defect backgrounds also creates great difficulties in their accurate identification. In addition, there are certain imbalances in the positive and negative samples of the fuselage dataset, making it unsuitable for direct application of existing semisupervised learning methods during training.

Second, current mainstream semisupervised learning approaches adopt the method that teachers and students learn from each other [8], [9]. Due to the imbalanced data distribution of classes, it is extremely difficult to generate an unbiased pseudolabel that matches the GT label with consistency using a single fixed confidence threshold. As shown in Fig. 1, in some categories, the number of pseudolabels is considerably higher than the GT label, while in other categories, the number of pseudolabels is much lower than the GT label, resulting in the quantity and quality of pseudolabels generated by the teacher model fluctuates highly during the whole training process, and unqualified pseudolabels will mislead the model to update. To alleviate this problem, some studies [8], [10] refine pseudolabels several times, while others [11] use feature maps as soft pseudolabels to avoid bias caused by nonmaximal suppression. If the above methods are directly applied to the semisupervised defect detection model for the fuselage, the inconsistency of the pseudolabels will be exacerbated. There are two main reasons: 1) it can be seen from Fig. 1 that in the four defect categories, the ratio of the number of

pseudolabels to the number of GT labels changes significantly. Compared with the general object detection dataset, the unevenness of data distribution and the sparsity of data will lead to more serious inconsistencies in the process of pseudolabel assignment. 2) Compared with the general object detection dataset, the aircraft fuselage skin dataset contains noise, outliers, low-quality images, and other problems. In the process of pseudolabel assignment, the assignment strategy based on a single threshold is easily affected by noise or outliers, resulting in the inconsistency of pseudolabels.

To address the aforementioned challenges, in this article, we aim to develop a scalable and efficient semisupervised industrial defect detection framework based on one-stage detector, considering both high accuracy and inference speed. We introduce the dynamic decoupled detection head and design a different representative basic defect detection model based on one-stage detector, called a Dynamic decoupled DETector (DyDET), which can effectively solve the first problem mentioned above. The issue of distribution mismatch of label data is mitigated by adopting fine-grained pseudolabel assignment strategies on different task branches and introducing class-aware adaptive pseudolabel assignment (CAPLA) strategies. By adaptively adjusting the pseudolabel assignment threshold, pseudolabels can be dynamically divided into reliable and uncertain regions, and different loss calculations are applied to pseudolabels in these different regions. The second problem mentioned above can be solved by introducing CAPLA strategies.

In summary, our main contributions are summarized as follows.

- 1) We introduced dynamic attention [12] and partial convolution [13] to address the problem of detecting small defects in aircraft fuselage and make the model lightweight. With dynamic attention, we can pay attention to the feature maps of small defects at different scales, to improve the detection accuracy of our model for small defects. By introducing partial convolution, we can reduce feature mapping redundancy in the channel.
- 2) We proposed a CAPLA strategy to adjust adaptively the threshold during the pseudolabel assignment process, which can solve the problem of label data distribution mismatch. By leveraging both labeled and unlabeled data, the CAPLA strategy addresses the challenge of limited labeled data availability to improve the accuracy and efficiency of defect detection in aircraft quality inspection.
- 3) On this basis, we constructed a quick and effective semisupervised defect detection framework based on the DyDET. In addition, we constructed an aircraft fuselage defect dataset named Aircraft_Fuselage_DET2023 for semisupervised training, which involved meticulously collecting and annotating a diverse range of images showcasing various types of defects. To evaluate the effectiveness of our proposed approach, we conducted an extensive series of experiments on this sub-dataset. The experimental results were highly promising, demonstrating that our proposed network achieved state-of-the-art performance while consuming fewer FLOPs than previous methods in detecting fuselage defects.

II. RELATED WORK

A. Semisupervised Object Detection

Semisupervised object detection has evolved since the approach to semisupervised image classification [14], [15], [16], [17], and there are two primary schemes: consistency based [18], [19] and pseudolabeling based [8], [9], [20], [21]. The latter has become the dominant solution in the field of semisupervised object detection. A simple semi-supervised learning framework for object detection (STAC) [20] uses data augmentation methods to process unlabeled data. To balance the effect of pseudolabel on the model, Soft Teacher [9] uses the score of pseudolabel as the weight computed by the loss function. Dense learning based semi-supervised object detection (DSL) [22] is the first attempt at semisupervised object detection using anchor-free detectors. Omni-supervised object detection with transformers (Omni-DETR) [23] implements a Transformer-based end-to-end unified framework for semisupervised object detection, under which different types of weak labels can be used to generate accurate pseudolabels through a bidirectional matching-based filtering mechanism for model learning. To alleviate the problem of inconsistent label distribution, Label-Match [10] uses the label data distribution to dynamically determine the threshold for filtering out pseudolabels during training. The above models perform well in terms of performance, but they do not show the best performance as one-stage object detectors commonly used in industrial surface defect detection. Therefore, our model is based on the semisupervised industrial surface defect detection framework proposed by YOLOV7 [24].

B. Defect Detection in Semisupervised Domain

He et al. [25] proposed a semisupervised GAN network-based method for steel surface defect classification. In the designed CAEGAN defect detection network, a CAE-based encoder is used and fed into a softmax layer to form the discriminator. Instead of directly predicting the true–false binary classification of the input image, the discriminator predicts $N + 1$ classes, where N denotes the number of defect types, and the additional classes indicate whether the input image is from the real dataset or a generator. He et al. [26] proposed a multitasking semisupervised learning method for steel surface defect classification, which uses cDCGAN to generate a large number of unlabeled samples. To take advantage of these unlabeled samples, the proposed model uses the multitasking fusion algorithm of cDCGAN and Resnet-18 for category label prediction on unlabeled samples. Gao et al. [27] also proposed a semisupervised learning method using convolutional neural networks to classify defects on steel surfaces and improved the performance of CNN by employing pseudolabels. Currently, semisupervised methods are mostly used for defect classification or recognition tasks but have not been widely used for localization and segmentation tasks.

C. Pseudolabel Assignment

Semisupervised object detection based on pseudolabel assignment is currently the dominant scheme, where label assignment is the key ingredient to determine the performance

of the detection model. Several studies [10], [28] have shown that the label assignment mechanism in supervised object detection cannot be directly used in semisupervised object detection, which leads to degradation of detection performance. Therefore, numerous efforts have been proposed to improve the label assignment mechanism in semisupervised object detection, such as ATSS [29], AutoAssign [30], and OTA [31]. In this article, based on the existing research, we propose a novel CAPLA strategy, which can effectively improve the accuracy of pseudolabel.

III. SEMISUPERVISED AIRCRAFT FUSELAGE DEFECT DETECTION NETWORK

In this section, we will provide an explanation of the proposed network for detecting defects in aircraft fuselage using a semisupervised approach. Specifically, we will present the overall pipeline in Section III-A. Following that, Section III-B will delve into the structure of the DyDET while Section III-C will elaborate on CAPLA. Last, in Section III-D, we will discuss the loss function employed by our model.

A. Overall Pipeline

In this section, we give an overview of the proposed aircraft fuselage defect detection network model, which is depicted in Fig. 2. Following previous work [8], [9], [10], [22], this framework is based on the student–teacher mutual learning method, and we use the DyDET as the base model due to its excellent detection accuracy and ability to meet real-time detection requirements. It is important to emphasize that our detection framework is not dependent on a specific basis model structure. Therefore, alternative models like a two-stage object detector or an anchor-free detector can be utilized as substitutes for this basis model. Throughout the structure of our model, our proposed DyDET effectively captures tiny defect features and enhances the model’s ability to generalize. The CAPLA strategy enables us to categorize these pseudolabels into reliable and uncertain ones based on a threshold that determines when to update them during training epochs. Reliable pseudolabels are utilized for default supervised training, while uncertain pseudolabels guide the training of student models. During the training procedure, the teacher model was enhanced through the utilization of exponential moving average (EMA) technology.

B. DyDET

In the detection of aircraft fuselage defects, the detection of small defects has always been a problem that needs attention and is expected to be effectively solved. A schematic representation of DyDET is depicted in Fig. 3. Inspired by dynamic head [12], we reconstructed the detection head of YOLOV7 [24] with three different kinds of attention. As shown in the dynamic head block in Fig. 3, assuming the input image $X \in \mathbb{R}^{640 \times 640 \times 3}$, after some feature extraction and fusion operations, the feature vector sequence $\mathcal{X}_{in} = \{X_1 \in \mathbb{R}^{20 \times 20 \times 1024}, X_2 \in \mathbb{R}^{40 \times 40 \times 512}, X_3 \in \mathbb{R}^{80 \times 80 \times 256}\}$ from different levels of the feature pyramid can be obtained. We have the option to employ either an up-sampling or down-sampling method to modify the dimensionality of the

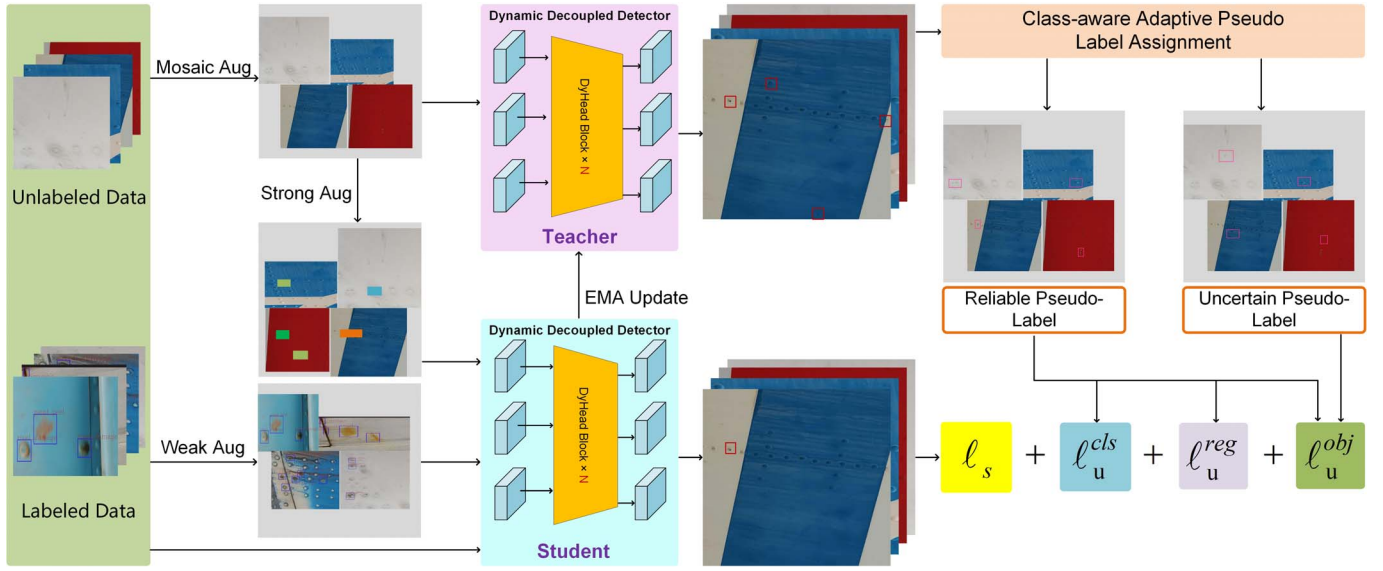


Fig. 2. Overview of a scalable and efficient semisupervised aircraft fuselage defect detection network. We propose two modules to implement this framework, namely DyDETs for improved location detection of tiny defects with enhanced inference efficiency and a CAPLA strategy that categorizes pseudolabels into two groups to address the issue of label data distribution mismatch.

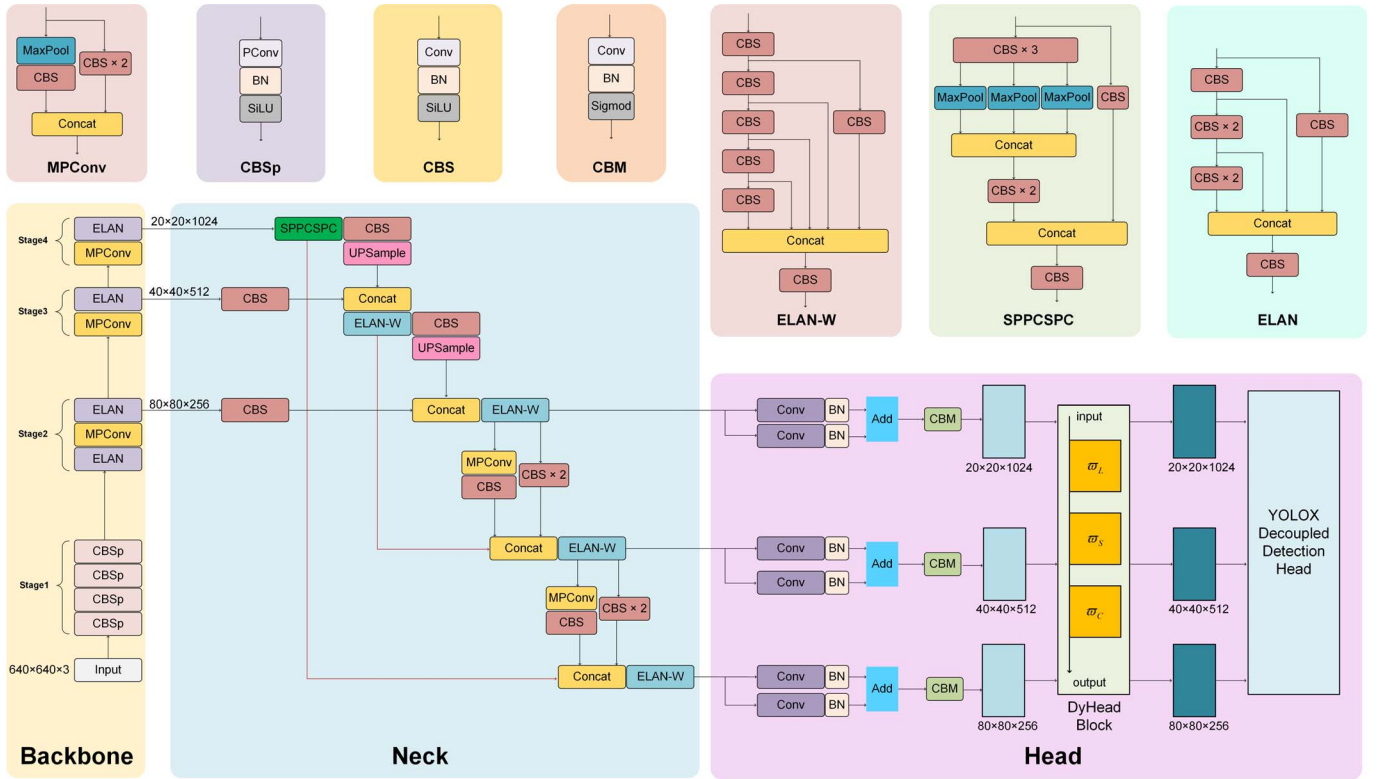


Fig. 3. Architecture diagram of DyDET based on YOLOV7 [24], which extends dynamic head block [12], and YOLOX decoupled detection head [32]. In addition, partial convolution [13] is introduced in the feature extraction module to reduce computational redundancy and memory access in DyDET. Among them, MPCConv, CBSp, CBS, CBM, ELAN-W, SPPCSPC, and ELAN represent different modules. The top part of the figure is a detailed description of each module, Conv refers to the convolution layer, Pconv refers to the partial convolution layer, BN refers to batch normalization, and Sigmoid and SiLU are activation functions.

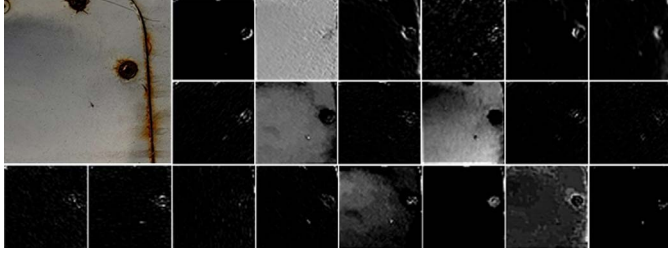


Fig. 4. Visualization of feature maps in an intermediate layer of a pretrained ResNet50, with the top-left image as the input. From a qualitative perspective, we can see the high redundancies across different channels.

feature vector, thereby aligning it with the scale of the intermediate feature. The rescaled feature vector can be expressed as a 4-D tensor $\mathcal{X} \in \mathcal{R}^{3 \times 40 \times 40 \times 512}$, where 3 is the count of feature vector in the sequence (i.e., the feature vector obtained from various scales), 40, 40, and 512 correspond to the dimensions of height, width, and channels of an intermediate feature. Then, we reshape the tensor into a 3-D tensor $\mathcal{X} \in \mathcal{R}^{3 \times 1600 \times 512}$. The resulting feature vector $\mathcal{X}$ is passed through three consecutive attentions in the dynamic head block in Fig. 3, each focusing exclusively on a single dimension of the feature vector. The attention function can be mathematically described as follows:

$$W(\mathcal{F}) = \varpi_C(\varpi_S(\varpi_L(\mathcal{X}) \cdot \mathcal{X}) \cdot \mathcal{X}) \cdot \mathcal{X} \quad (1)$$

where $\varpi_L(\cdot)$, $\varpi_S(\cdot)$, and $\varpi_C(\cdot)$ represent three distinct attention functions applied to dimensions L , S , and C , correspondingly.

In addition, to resolve the conflict between the classification task and the regression task, we add a decoupled detection head [32] to the DyDET. By adding decoupled heads to the DyDET, we can further distinguish between the classification and localization tasks of defects, thereby improving the classification and localization capabilities of our model. As a defect detection model, we have been trying to design an efficient and lightweight model without losing the accuracy of the model detection. By analyzing our Aircraft_Fuselage_DET2023 dataset, we found that the feature maps between different channels have a high degree of similarity, as shown in Fig. 4. Therefore, the use of feature mapping redundancy can make our model lightweight. Specifically, we introduce a partial convolution (Pconv) [13] to reduce both computational redundancy and memory access for dynamic decoupling detectors. It only applies the conventional convolution operation to a part of the input channels to extract the spatial features, and the rest of the channels remains unchanged employing residual connection. Notably, Pconv [13] keeps the remaining channels unchanged instead of feature mapping, an operation that allows feature information to flow between channels and enhances the interaction of spatial feature information.

C. CAPLA

The key issue in pseudolabel-based semisupervised object detection is how to efficiently assign pseudolabels. Mismatch in the distribution of label data can lead to suboptimal assignment

of pseudolabels and deterioration of the mutual learning mechanism. Pseudolabel filter is a common pseudolabel assignment strategy. It simply sets a threshold, filters out pseudolabels below that threshold, and treats them as background. However, this strategy often leads to suboptimal assignment, as shown in Fig. 5. In the upper part of Fig. 5, the pseudolabels in the blue boxes will be filtered out as background since their threshold value is lower than the set threshold, while the pseudolabels in the red boxes will be utilized for training the student model. However, this simplistic assignment method may not apply to defect categories whose label data mismatch. As shown in Fig. 1, some categories (such as defect type of scratch) generate a relatively large number of pseudolabels, so it is very possible to filter out more reliable pseudolabels; while in other categories (such as defect type of rivet-damage) generate a small number of pseudolabels, the quality of which is poor. The pseudolabel used to update the student model is of poor quality, resulting in a sharp decline in the performance of the student model. In Fig. 5 (below), our proposed CAPLA strategy is illustrated. In our work, a CAPLA strategy provides a finer assignment of pseudolabels generated by dynamically decoupled detectors. Inspired by the work of Label Match [10], we obtain the adaptive threshold by minimizing the difference in distribution between labeled data and unlabeled data, whose expression is as follows:

$$\begin{aligned} & \underset{t_1, \dots, t_C}{\operatorname{argmin}} D_{\text{KL}}(\underbrace{[r_1^l, \dots, r_C^l]}_{f-f}, \underbrace{[r_f^l]}_{f-b}, \underbrace{[r_1^u, \dots, r_C^u]}_{f-f}, \underbrace{[r_f^u]}_{f-b}) \\ & \text{s.t. } r_C^l = \frac{n_c^l}{\sum_{i=1}^C n_i^l} \\ & \quad r_f^l = \frac{\sum_{i=1}^C n_i^l}{N_l} \\ & \quad r_C^u = \frac{\sum_{j=1}^{N_u} \sum (P_i^j > t_c)}{\sum_{i=1}^C \left(\sum_{j=1}^{N_u} \sum (P_i^j > t_c) \right)} \\ & \quad r_f^u = \frac{\sum_{i=1}^C \left(\sum_{j=1}^{N_u} \sum (P_i^j > t_c) \right)}{N_u} \end{aligned} \quad (2)$$

where $D_{\text{KL}}(\cdot, \cdot)$ represents the Kullback–Leibler divergence between the distribution of label data and unlabeled data, n_i^l denotes the box number of the i th class in the label data, C is the entire foreground class number (in this article, $C = 4$), P_i^j is a list of prediction scores of the i th class in the j th unlabeled image, $f - f$ refers to the foreground–foreground class distribution, $f - b$ indicates the foreground–background class distribution, and N_u and N_l denote the number of labeled and unlabeled data. The confidence threshold t_c for filtering pseudoboxes for the c th category is determined as follows:

$$t_c = P_c^l \left[\delta\% \cdot n_c^l \cdot \frac{N_u}{N_l} \right] \quad (3)$$

where P_c^l is a list of pseudolabel scores of the c th class at the l th epoch, which are sorted by descending, $\delta\%$ represents the proportion of predefined reliable pseudolabels, and n_c^l represents the number of c th class GT labels at the l th epoch.

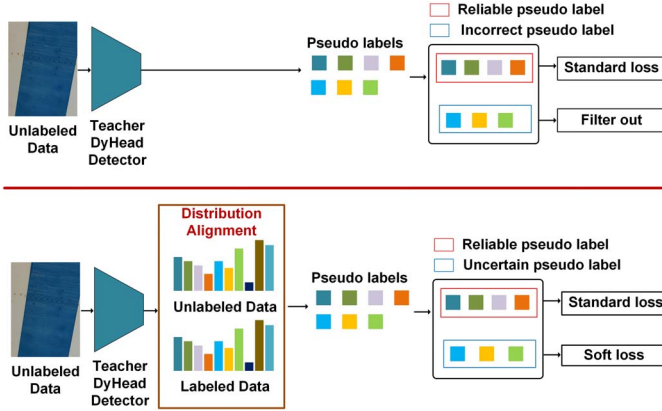


Fig. 5. Illustrations of different pseudolabels assignment strategies. The upper part of the figure is pseudolabel filter [8], which filters pseudolabels by setting a single threshold, which may lead to suboptimal assignment, thus negatively affecting network training. In the lower part of the figure, we proposed an adaptive pseudolabel allocation method based on class awareness, which divides pseudolabels into reliable and uncertain regions by obtaining adaptive thresholds. The pseudolabels in different regions are optimized with different loss functions in the training process.

It is important to note that the quality of pseudolabels cannot be guaranteed, especially during early training. To ensure effective implementation, a limited portion of unlabeled data is chosen to estimate the distribution utilized for determining the threshold. While the training process involves consecutive optimization of models, it is observed that the previous thresholds are not accurate enough for pseudolabels and do not align with the actual distribution of classes. Thus, we periodically revise these thresholds every $\mathcal{K}$ iterations to adaptively accommodate the current teacher model. In this way, the threshold is dynamically adjusted for each class and adaptively up-to-date, termed a CAPLA, which we consider a critical step in addressing the issue of the fuselage defect label data distribution mismatch.

D. Loss Function

The loss of a semisupervised aircraft fuselage defect detection network is defined as a pair of single labeled images and an unlabeled image

$$\ell = \ell_s + \lambda \ell_u \quad (4)$$

where ℓ_s represents the loss function computed on a labeled image, while ℓ_u represents the loss function computed on an unlabeled image, λ is used to balance the supervised loss and the semisupervised loss, which is set to 2.0 in this article. The ℓ_s is defined as follows:

$$\ell_s = \sum_{h,w} \left(\text{CE} \left(X_{(h,w)}^{\text{cls}}, Y_{(h,w)}^{\text{cls}} \right) + \text{FocalIoU} \left(X_{(h,w)}^{\text{reg}}, Y_{(h,w)}^{\text{reg}} \right) + \text{CE} \left(X_{(h,w)}^{\text{obj}}, Y_{(h,w)}^{\text{obj}} \right) \right) \quad (5)$$

where $\text{CE}(\cdot, \cdot)$ indicates cross-entropy loss function, $X_{(h,w)}$ is the output of the student model, and $Y_{(h,w)}$ means the sampled results generated by the label assignment of DyDET. In addition, superscripts “cls,” “reg,” and “obj” represent classification

TABLE I
NUMBER OF EACH CATEGORY IN
THE ORIGINAL AIRCRAFT
FUSELAGE DEFECT
DATASET

Type	Count	Total
Paint-peel	1400	5990
Rivet-damage	800	
Rust	1000	
Scratch	2790	

score at position (h, w) , regression score at position (h, w) , and objectness score at position (h, w) , respectively. For fuselage defect data, it is necessary to consider that category imbalance may lead to inaccurate regression of box, thus affecting the detection of tiny defects. FocalIoU($\cdot, \cdot$) is a variant of IoU loss that can not only take into account the width and height of the regression box and the distance between the center of the real box and the prediction box but also solve the problem of category imbalance of the fuselage defect dataset to a certain extent. The ℓ_u is defined as follows:

$$\ell_u = \ell_u^{\text{cls}} + \ell_u^{\text{reg}} + \ell_u^{\text{obj}} \quad (6)$$

$$\ell_u^{\text{cls}} = \sum_{h,w} \left(\mathbb{I}_{\{p_{(h,w)} > t_c\}} \text{CE} \left(X_{(h,w)}^{\text{cls}}, \hat{Y}_{(h,w)}^{\text{cls}} \right) \right) \quad (7)$$

$$\ell_u^{\text{reg}} = \sum_{h,w} \left(\mathbb{I}_{\{p_{(h,w)} > t_c\}} \text{FocalIoU} \left(X_{(h,w)}^{\text{reg}}, Y_{(h,w)}^{\text{reg}} \right) \right) \quad (8)$$

$$\ell_u^{\text{obj}} = \sum_{h,w} \left(\mathbb{I}_{\{p_{(h,w)} \leq t_c\}} \text{CE} \left(X_{(h,w)}^{\text{obj}}, \mathbf{0} \right) + \mathbb{I}_{\{p_{(h,w)} > t_c\}} \text{CE} \left(X_{(h,w)}^{\text{obj}}, \hat{Y}_{(h,w)}^{\text{obj}} \right) \right) \quad (9)$$

where $\hat{Y}_{(h,w)}^{\text{cls}}$, $\hat{Y}_{(h,w)}^{\text{reg}}$, and $\hat{Y}_{(h,w)}^{\text{obj}}$ are the classification score, regression, and objectness score of sampled results from class-aware adaptive pseudolabels assignment at location (h, w) on feature map separately. $p_{(h,w)}$ is the score of pseudolabel at (h, w) . $\mathbb{I}_{\{\cdot\}}$ is the indicator function, which outputs 1 if condition $\{\cdot\}$ is satisfied and 0 otherwise.

IV. EXPERIMENTS

A. Datasets

We constructed an aircraft fuselage defect detection dataset named Aircraft_Fuselage_DET2023, which can be accessed via <https://doi.org/10.21227/3ref-ex71>, to verify the effectiveness of the proposed model. We used cameras to photograph and document surface damage to the aircraft fuselage, including scratch, paint-peel, rivet-damage, and rust. After data screening, filtering, data addition, and balancing the number of categories, there are 5601 original pictures of the aircraft fuselage defect dataset in this article, and the number of each category in the original dataset is shown in Table I. During dataset labeling, 389 defect images were carefully selected for labeling, and the remaining 5212 images were used as unlabeled datasets in the experiments. To better train and validate the models, these images

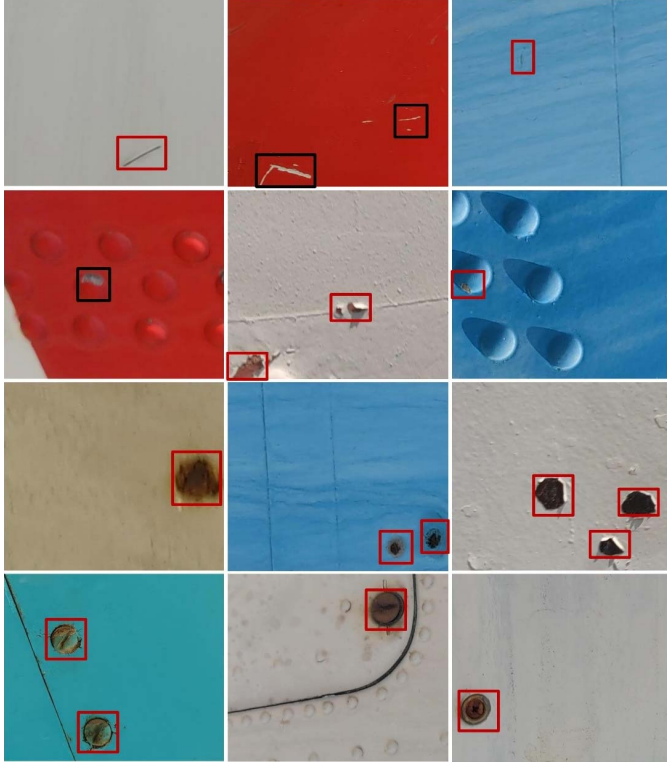


Fig. 6. Defect images on aircraft fuselage dataset. Row1: scratch; row2: paint-peel; row3: rust; row4: rivet-damage. From these images, we can see that the challenges of defect inspection include tiny defects (the image pixel size is 1120×960 , and the average pixel size of the defect is 80×68), with various background disturbances for the same defect type, defects in the same class that are really different.

are augmented with basic data augmentation such as horizontal and vertical flips. After some basic data augmentation, the total number of labeled datasets is 3112. Typical images for each category are shown in Fig. 6. The type of defect in the first row of Fig. 6 is scratch, which is primarily caused by the use of metal tools or improperly rough tools while removing sealant from the edge of the fuselage or adhesive tags from the outer surface of the in-position cutting machine. The diversity of their shapes and sizes poses a great challenge for detection. In the second row of Fig. 6, the defect type is paint-peel, which has color shift and rough edges, mainly caused by destructive oxidation and a high-temperature environment. The greatest detection difficulty for this type of defect is the large background interference. The defect type in the third row of Fig. 6 is rust, which is mainly caused by rainy weather and humid environment. Most of these imperfections are very minute and easily confused with the background color. The defect type in the fourth row of Fig. 6 is rivet-damage, which may result from loosening, falling off, corrosion, and other damage caused by long-term exposure to alternating load, corrosive environment, and vibration environment. Currently, due to the limitation of the dataset of aircraft skin surface defects, the models and comparison models presented in this article only target the four types of defects mentioned above. Further types of surface defects will arise in future practical applications. The proposed method is also applicable after training as long as sufficient datasets are available as support.

B. Implementation Details

The hardware configuration of algorithm environment: Central processing unit (CPU): Intel(R) Core(TM) i9-9920X CPU @3.50 GHz; Graphics Processor (GPU): NVIDIA GeForce RTX 2080Ti GPU, 128 GB of memory. The software configuration is PyTorch development framework [33], Python programming language. For the training configuration, the batch size is 16 for labeled data and 16 for unlabeled data and the learning rate is 0.01. The EMA smoothing hyperparameter is 0.995. We used weak and strong data augmentation. In data augmentation, Mosaic, flip left and right, large-scale dithering, grayscale, Gaussian blurring, and color space conversion were selected. In addition, to avoid overfitting the model, we visualized the training process. The loss curve and the metric curve are shown in Figs. 7 and 8. As seen from the above figures, the model converges when the number of epochs reaches 400. Therefore, we set the maximum number of epochs is 400.

C. Experimental Results and Analysis

In this section, we evaluate our proposed model on a dataset of aircraft fuselage defects. We select a variety of mainstream semisupervised object detection models with excellent performance in the semisupervised object detection domain, including the Transformer-based semisupervised object detection model, to show the superiority of our model in aircraft fuselage defect detection. The metrics we used were mAP_{50} and $mAP_{50:95}$, and we report comparisons with other semisupervised object detection models in Table II. All results are the average fivefolds. 10%, 25%, 50%, and 100% refer to the data sampled proportionally from 3112 labeled data as labeled data in the training process, and the rest are 8000 unlabeled data. Under the same data configuration, it can be seen from Table II that the model trained under the proposed framework is superior to other models, and the FLOPS of the model proposed by us is smaller. In particular, when we select all the label data, the overall performance of the model improves significantly. In the setting of semisupervised object detection, the quality of pseudolabels plays a decisive role in the detection performance. Therefore, a certain amount of label data must be used to guide the generation of pseudolabels. As can be seen from Table II, when we select 10% and 25% of the labeled data, respectively, the performance of the model is mediocre.

Compared to Label Match [10], which is a model to solve the problem of inconsistent label assignment, our model can improve the performance by 1.1(10%), 0.7(25%), 3.7(50%), and 1.1(100%) under the metric $mAP_{50:95}$, respectively. Compared with the semisupervised object detection framework based on Transformer, Omni-DETR [23], our proposed approach continuously improves the performance by 5.2(10%), 5.9(25%), 14.4(50%), and 7.3(100%) under the metric mAP_{50} , respectively. In addition, compare to the latest state-of-the-art work, dense teacher guidance for semi-supervised object detection (DTG-SSOD) [34], our method can improve the performance by 0.5(10%), 1.3(25%), 6.1(50%), and 3.6(100%) under the metric $mAP_{50:95}$, respectively.

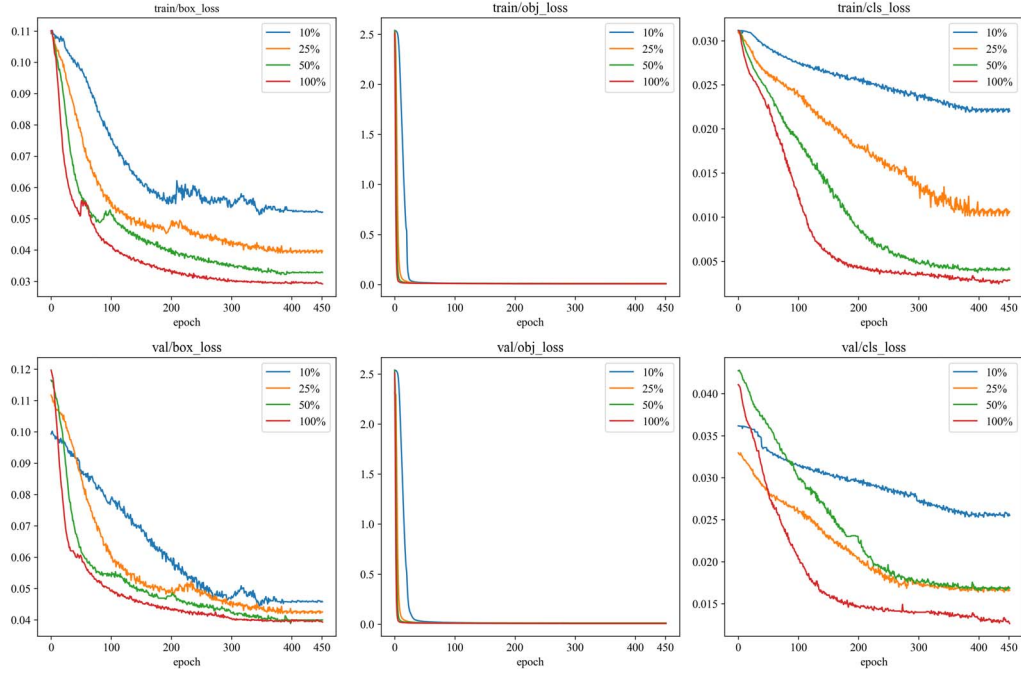


Fig. 7. Visualization of loss curves. The top half of the figure is the training loss curve, and the bottom of the figure is the validation loss curve. box_loss, obj_loss, and cls_loss represent regression loss, objectness loss, and classification loss, respectively. 10%, 25%, 50%, and 100% refer to the data sampled proportionally from 3112 labeled data as labeled data in the training process.

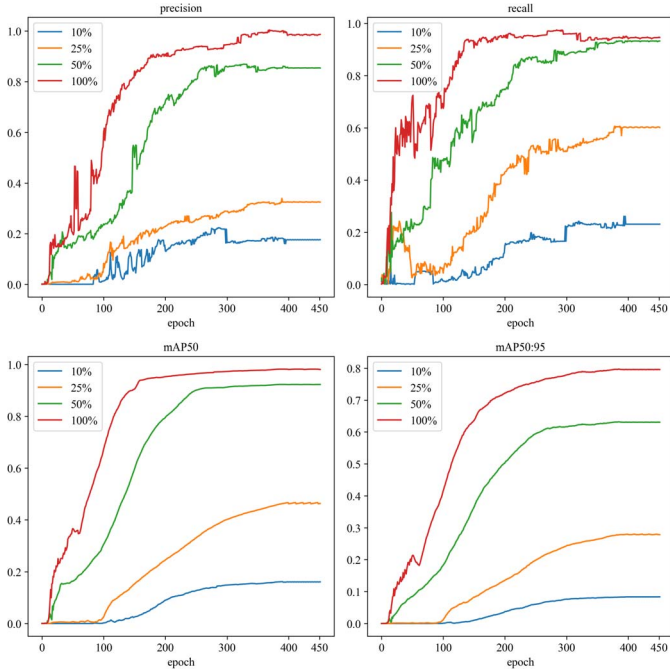


Fig. 8. Visualization of metric curves. 10%, 25%, 50%, and 100% refers to the data sampled proportionally from 3112 labeled data as labeled data in the training process.

V. DISCUSSION AND ANALYSIS

A. Effect of Dynamic Decoupled Detector

1) *Comparison With State-of-the-Art Methods:* To comprehensively evaluate the performance of our proposed DyDET,

we conducted a series of experiments on aircraft fuselage defect dataset using YOLOV4 [37], YOLOV5L, YOLOV7 [24], and DyDET. The detection performance of these state-of-the-art models was compared to our dataset, and the results are presented in Fig. 9. In the experiments, we trained the model using fuselage labeled dataset, and the rest of the training configuration and testing remained the same. Through comparative analysis, it was clear that our proposed DyDET outperformed other methods in terms of efficiency and accuracy. This was primarily due to the innovative dynamic attention mechanism employed in the DyDET, which allows for more efficient processing of complex fuselage defect images and improved detection capabilities.

2) *Ablation Studies:* We conducted a series of ablation experiments to verify the effectiveness and efficiency of the dynamic attention block. First, we conducted a controlled study of effectiveness in dynamic head blocks by gradually adding different attention modules to the baseline. As shown in Table III, ϖ_L , ϖ_S , and ϖ_C represent scale-aware attention module, spatial-aware attention module, and task-aware attention module, respectively. It is worth noting that individually adding each attention module to the baseline improves performance by 0.4, 1.7, and 1.3 under the metric mAP₅₀, respectively. In addition, we can observe that the spatial-aware attention module achieves the biggest gain because for defect detection spatial location dominates. When we add both ϖ_L and ϖ_S to the baseline, it continuously improves the performance by 1.8 under the metric mAP₅₀. We add three attention modules to the baseline that can improve the performance by 2.5 under the metric mAP₅₀. These results demonstrate that different components work as a coherent module. Second, we conducted a

TABLE II
 COMPARISON OF MODEL PERFORMANCE (MAP₅₀ AND MAP_{50:95})

Method	mAP ₅₀				mAP _{50:95}				FLOPS
	10%	25%	50%	100%	10%	25%	50%	100%	
STAC [20]	13.0 ± 0.42	41.4 ± 0.04	76.5 ± 0.33	86.7 ± 0.10	3.1 ± 0.35	22.8 ± 0.12	53.9 ± 0.12	73.8 ± 0.22	202.13G
Instant-teaching [21]	12.5 ± 0.14	43.6 ± 0.39	79.5 ± 0.29	88.3 ± 0.05	4.7 ± 0.15	24.6 ± 0.14	56.4 ± 0.14	75.9 ± 0.04	202.13G
Soft teacher [9]	14.2 ± 0.22	44.9 ± 0.25	80.3 ± 0.51	90.7 ± 0.09	7.1 ± 0.39	26.7 ± 0.12	57.1 ± 0.21	77.3 ± 0.13	204.12G
Label Match [10]	13.4 ± 0.19	44.1 ± 0.19	84.6 ± 0.30	93.5 ± 0.13	7.3 ± 0.28	27.2 ± 0.13	59.3 ± 0.11	78.4 ± 0.35	202.12G
Unbiased teacherv2 [28]	12.8 ± 0.09	43.3 ± 0.11	80.7 ± 0.19	90.2 ± 0.32	6.9 ± 0.42	25.8 ± 0.12	56.1 ± 0.08	76.3 ± 0.24	200.59G
Activate Teacher [35]	10.4 ± 0.20	43.5 ± 0.11	83.3 ± 0.14	92.4 ± 0.47	7.9 ± 0.13	26.8 ± 0.05	58.6 ± 0.21	77.5 ± 0.16	204.12G
Dense Teacher [11]	11.5 ± 0.54	42.0 ± 0.41	81.5 ± 0.20	89.0 ± 0.24	8.0 ± 0.31	26.0 ± 0.12	57.8 ± 0.05	76.7 ± 0.04	200.59G
DTG-SSOD [34]	10.7 ± 0.06	42.3 ± 0.32	78.8 ± 0.22	90.8 ± 0.15	7.9 ± 0.12	26.6 ± 0.11	56.9 ± 0.52	75.9 ± 0.30	200.59G
DSL [22]	14.6 ± 0.26	41.9 ± 0.03	77.5 ± 0.41	92.4 ± 0.21	8.1 ± 0.28	26.1 ± 0.15	57.2 ± 0.10	76.5 ± 0.30	200.59G
MUM [36]	12.4 ± 0.17	41.1 ± 0.34	76.9 ± 0.33	86.7 ± 0.32	7.8 ± 0.12	25.4 ± 0.15	56.5 ± 0.28	74.6 ± 0.32	200.59G
Omni-DETR [23]	10.9 ± 0.21	40.8 ± 0.15	77.8 ± 0.14	90.8 ± 0.36	6.7 ± 0.21	24.9 ± 0.13	57.0 ± 0.14	74.9 ± 0.18	208.12G
Our method	16.1 ± 0.15	46.7 ± 0.30	92.2 ± 0.10	98.1 ± 0.22	8.4 ± 0.13	27.9 ± 0.21	63.0 ± 0.15	79.5 ± 0.32	179.02G

Note: All results are the average of fivefolds. 10%, 25%, 50%, and 100% refer to the data sampled proportionally from 3112 labeled data as labeled data in the training process. Bold indicates the best results.

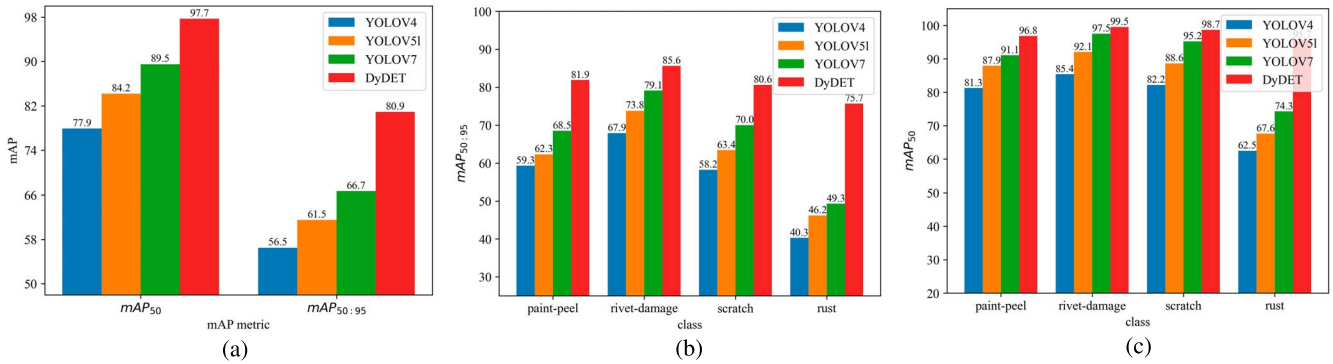


Fig. 9. Ablation study about (a) performance comparison of different models on fuselage defect datasets. (b) Comparison of detection accuracy of different defect types (mAP_{50:95}). (c) Comparison of detection accuracy of different defect types (mAP₅₀).

 TABLE III
 ABLATION STUDY ON THE EFFECTIVENESS OF EACH ATTENTION MODULE IN THE DYNAMIC HEAD BLOCK

ϖ_L	ϖ_S	ϖ_C	mAP ₅₀	mAP _{50:95}
×	×	×	89.5	66.7
✓	×	×	89.9	67.0
×	✓	×	91.2	69.1
×	×	✓	90.8	68.4
×	✓	✓	92.5	70.1
✓	×	✓	91.8	69.9
✓	✓	×	92.3	70.3
✓	✓	✓	93.0	70.9

Note: Bold indicates the best results.

 TABLE IV
 ABLATION STUDY ON THE EFFICIENCY AND EFFECTIVENESS OF STACKING DIFFERENT NUMBERS OF DYNAMIC HEAD BLOCKS

Dyhead Blocks	mAP ₅₀	mAP _{50:95}	GFLOPs
Baseline	89.5	66.7	105.2G
1	93.0	70.9	101.1G
2	94.6	72.5	102.4G
4	97.7	80.9	104.9G
6	96.9	79.4	107.4G
8	95.8	78.2	109.9G

Note: Bold indicates the best results.

series of experiments to evaluate the efficiency of the dynamic attention head, we controlled the depth (number of blocks) and compared their performance and cost of computation (GFLOPs)

to the baseline, as shown in Table IV. By varying the number of dynamic attention blocks used (e.g., 1, 2, 4, 6, and 8), we found that the dynamic attention head can benefit from an increase in depth by stacking more blocks. Interestingly, the one-block approach already outperforms the baseline at a

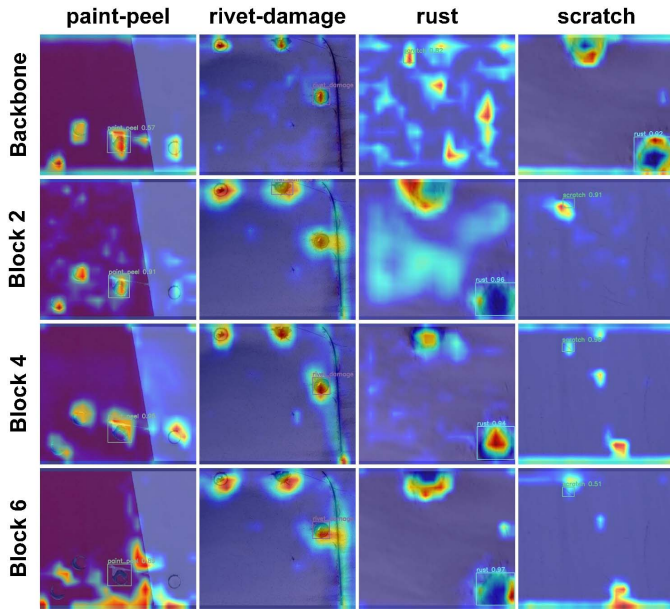


Fig. 10. Visualization of the effectiveness of different numbers of the dynamic attention head block.

lower computational cost. Moreover, even with four blocks, the computational cost is still lower than the baseline compared to the computational cost of the backbone, while greatly improving the accuracy. This highlights the efficient nature of the dynamic attention head design, as it enables us to achieve better performance with fewer resources.

3) *Analysis of Dynamic Attention Block:* There is a phenomenon worth noting in Table IV, which we can observe that when the number of blocks is 4, the performance improvement is most obvious. Still, if we continue to add blocks, the performance deteriorates. To further explain this phenomenon, we visualized the heat maps of different numbers of attention blocks as shown in Fig. 10, which shows the feature map output before and after applying different numbers of dynamic attention blocks, demonstrating the improvement in focusing on foreground defects. Before the application of the dynamic attention block, the feature maps of defects extracted from the backbone were noisy and could not effectively focus on foreground defects. As the feature map passes through more attention blocks, we can clearly see that the defect feature map covers more valid foreground defects and focuses more accurately on the space location of different class defects. However, an increase in the number of dynamic attention modules does not consistently lead to improved performance. From Fig. 10, we can observe that when the dynamic attention block is 6, the focus area of the model will be biased, and it cannot focus on the defect area. This is most likely because there is a lot of noise in the defect image itself, and an appropriate dynamic attention block can focus on the defect area better. However, with the increase of dynamic attention blocks, the attention will also overlearn the features of the nondefect region, which will lead to the confusion of defect features and nondefect features, and then reduce the detection performance of the model.

B. Effect of CAPLA

1) *Comparison With Other Pseudolabel Assignment Methods:* We compare different label assignment strategies for pseudolabels, which are widely used in semisupervised learning. The three strategies we examine are as follows: AutoAssign [30]; LabelMatch [10]; and Pseudolabel assigner [38]. For a fair comparison, we train the proposed semisupervised aircraft fuselage defect detection network with different pseudolabel assignment strategies described above. The performance of each strategy is evaluated using the dataset of aircraft fuselage defects. Fig. 11(c) shows the superiority of our proposed class-aware pseudolabel assignment strategy over other label assignment strategies. The CAPLA strategy takes into account the imbalance between the real label and the background and assigns pseudolabels dynamically based on the data distribution of pseudolabels. This approach helps to reduce the impact of background noise and label assignment errors on the student model's performance. For AutoAssign [30], there is an imbalance between the real label and the background, which leads to the background dominating the detection training. As a result, defects are often treated as the background after training, hindering performance improvement. For Pseudolabel assigner [38], a single threshold is used to forcibly divide pseudolabels into hard pseudolabels and pseudolabels, which can result in some pseudolabels assignment errors. This approach does not take into account the data distribution of pseudolabels, leading to poor performance of the student model, as shown in the upper part of Fig. 5. For LabelMatch [10], the threshold for dividing pseudolabels is also obtained dynamically by using the data distribution of pseudolabels. However, this approach does not apply the unreliable pseudolabels in training. It can be seen from Fig. 11(c) that compared with the previous three label assignment strategies, the CAPLA strategy proposed by us is more suitable for the distribution of pseudolabel data in aircraft fuselage defect dataset and has improved its performance to some extent.

2) *Ablation Studies:* In this experiment, we employed the same experimental setup and selected 100% labeled data to determine the update interval $\mathcal{K}$ for the CAPLA strategy. As described in Section III-B, our proposed CAPLA is updated into the teacher model during the training phase to avoid error bias due to outdated predictions. As can be observed from Fig. 11(a), different update intervals have varying impacts on the determination of adaptive thresholds, and the selection of update intervals must be determined. The update interval $\mathcal{K}$ represents the frequency at which the pseudolabels are updated, and it plays a significant role in determining the performance of our method. Furthermore, it is worth noting that the choice of update interval $\mathcal{K}$ depends on the size of the dataset. In general, a larger update interval $\mathcal{K}$ leads to more stable adaptive thresholds, but it may result in longer training time. Conversely, a smaller update interval $\mathcal{K}$ can reduce the training time, but it may lead to less stable adaptive thresholds. In our experiment, we just used $\mathcal{K} = 2000$ to update the threshold. In the training phase, we calculate the adaptive threshold based on the reliability score ratio. In Fig. 11(b), we analyze the effects of different

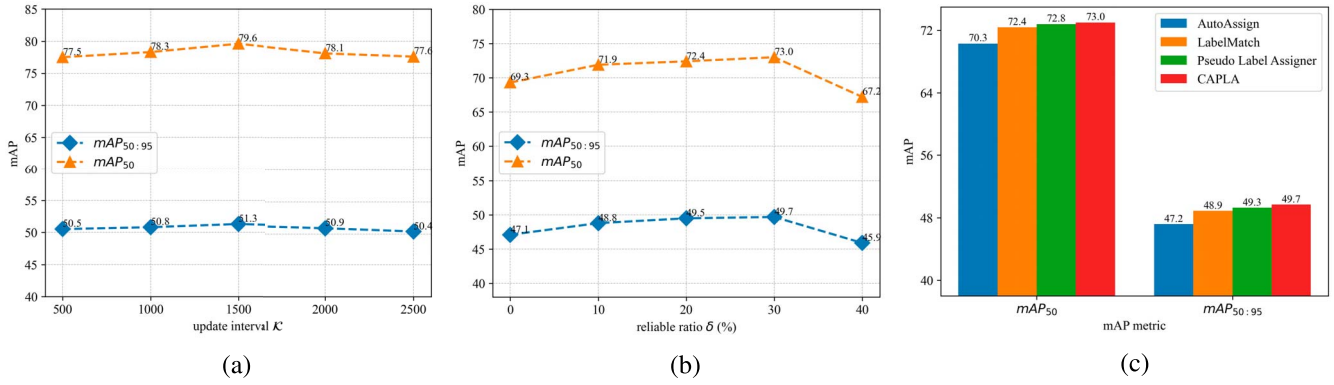

 Fig. 11. Ablation study about: (a) the updating interval K ; (b) the effect of reliable ratio δ ; and (c) different label assignment strategies.

 TABLE V
ANALYSIS OF THE BATCH SIZE. NOTE
THAT THE LEARNING RATE AT
THIS POINT IS 0.01

Batch Size	mAP_{50}	$mAP_{50:95}$
2	94.2	70.4
4	93.4	66.1
8	94.9	71.2
16	96.8	74.5

Note: Bold indicates the best results.

 TABLE VI
ANALYSIS OF THE LEARNING RATE.
NOTE THAT THE BATCH SIZE AT
THIS POINT IS 16

Learning Rate	mAP_{50}	$mAP_{50:95}$
0.001	56.3	40.4
0.01	97.6	74.9
0.1	70.5	43.4

Note: Bold indicates the best results.

ratios on the results. As shown in Fig. 11(b), if we directly set all candidate pseudolabels as uncertain pseudolabels ($\delta = 0$), the performance is inferior to classifying some pseudolabels as reliable pseudolabels, mainly due to the lack of box regression optimization for unlabeled data. However, having too many reliable pseudolabels is also detrimental to model updates, as pseudolabels will bring a certain amount of noise. To strike a balance between these two competing objectives, we conducted a series of experiments with different values of δ . We found that setting $\delta = 30$ yielded the most favorable results in terms of both performance and model updates. This value of δ allowed us to achieve a good tradeoff between the number of reliable pseudolabels and the noise they introduced into the training data. As a result, we conducted all the experiments with $\delta = 30$.

C. Analysis of Different Hyperparameters

By comparing the effect of different hyperparameters, we can choose the best hyperparameter. Therefore, for the hyperparameter batch size and learning rate, we carried out sufficient verification experiments. Table V shows the ablation experiments with different batch sizes. We selected batch size as 2, 4, 8, and 16, and the learning rate was set to 0.01. As can be seen from the results in Table V, the model performs optimally when the batch size is 16. In addition, Table VI shows the ablation experiments with different learning rates. We selected the learning rates as 0.001, 0.01, and 0.1, and the batch size was set to 16. From Table VI, we can choose the best learning rate is 0.01. Based on the above analysis, we finally select the hyperparameter batch size as 16 and the learning rate as 0.01.

 TABLE VII
COMPARISON OF MODEL PERFORMANCE UNDER
TWO LEARNING PARADIGMS (I.E., SUPERVISED
LEARNING AND SEMISUPERVISED LEARNING)

Learning Paradigm	mAP_{50}	$mAP_{50:95}$
Supervised Learning	89.5	66.7
Semisupervised Learning	95.2	72.5

D. Analysis of the Feasibility of Model

The challenge of aircraft fuselage defect detection based on deep learning is that the detection accuracy of small defects is not high, subject to noise and complex background interference, and which is an asset-heavy route for the data-driven supervised learning paradigm. Therefore, we compare the detection performance of the models under two different learning paradigms (i.e., supervised learning and semisupervised learning). Under the supervised learning paradigm, we use a DyDET to train on 3112 labeled datasets. Under the semisupervised learning paradigm, we use all labeled data as supervision and use all unlabeled data to train our model. The results under the two paradigms are shown in Table VII. From Table VII, we can observe that our model improves the performance by 8.2 and 10.8 under the metric mAP_{50} and $mAP_{50:95}$ in the semisupervised learning paradigm, respectively. This result proves that unlabeled data play an important role in improving the performance of the defect detection network and save the time and cost of manually labeling data. In addition, we visualized the final detection results of the model on the test set. As can be



Fig. 12. Visualization of the final detection results of our model.

seen from Fig. 12, despite the complex background diversity of aircraft fuselage defects the large variation of defect scale, and the majority of small defects, our detector can still detect these defects very well. This also proves that our proposed method can effectively solve the above problems.

However, although we have demonstrated the superiority of the proposed model, there are still two issues that cannot be ignored. First, while the detection accuracy of tiny defects has been greatly improved, it is still not at a reasonable level for industrial quality inspection. Further improvements and optimizations are needed to address this major problem in current aircraft fuselage defect detection. Second, our proposed CAPLA strategy is suitable for addressing the hypothesis that there exists a distribution mismatch between labeled and unlabeled data in the fuselage defect dataset. However, assuming that labeled and unlabeled data follow the same data distribution in other surface defect datasets renders our proposed method inapplicable. The above two main difficulties are our future research focus. For the detection of tiny defects, we will focus on the effective integration of global information and local semantic information. The global feature information can roughly locate the defect, while local semantic information can accurately detect tiny-scale defects. For the pseudolabel assignment strategy, we need to solve how to adaptively select different pseudolabel strategies according to the prior distribution of labeled data and unlabeled data, which will be interesting and challenging for future work.

VI. CONCLUSION

In this article, we propose a semisupervised network for aircraft fuselage defect detection that effectively leverages a large amount of unlabeled data to enhance model accuracy and generalization ability, while reducing the time and cost associated with manual labeling. To address the challenge of detecting

tiny defects on aircraft fuselage, we introduce a DyDET that integrates scale-aware attention, task-aware attention, and spatial-aware attention to significantly improve detection accuracy. Furthermore, through analysis of the aircraft fuselage defect dataset, we identify a label data distribution mismatch issue between labeled and unlabeled data which results in poor-quality pseudolabels during model training. To overcome this problem, we present a CAPLA strategy that dynamically adjusts the threshold for assigning pseudolabels during training to enhance the quality of generated labels for better student model updates. We conduct extensive experiments on an aircraft fuselage defect dataset to evaluate our proposed model's performance. The experimental results demonstrate superior performance compared to mainstream semisupervised learning models for object detection. These proposed methods are essential and effective in ensuring high-quality detection performance.

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